

## 2. Git-HOL

### What is .gitignore?

.gitignore is a special file in Git that tells Git **which files or folders should not be tracked**.

Sometimes a project contains files that should **not** be uploaded to GitHub, such as:

- Temporary files
- Log files
- Compiled files
- Environment files containing passwords or API keys
- IDE settings

Instead of deleting these files, we list them in the .gitignore file. Git then ignores them.

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### Why do we use .gitignore?

We use .gitignore to:

- Keep the GitHub repository clean.
- Avoid uploading unnecessary files.
- Protect sensitive information like passwords and API keys.
- Prevent machine-specific files from being shared.

For example:

- .env may contain database passwords.
- node\_modules contains thousands of installed packages that can be downloaded again using npm install, so it doesn't need to be pushed.

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### How to Ignore Unwanted Files Using .gitignore

### **Step 1: Create a file named**

`.gitignore`

in the root folder of your project.

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### **Step 2: Add the files or folders you want Git to ignore.**

Example:

# Ignore log files

`*.log`

# Ignore environment file

`.env`

# Ignore compiled files

`*.class`

# Ignore node\_modules folder

`node_modules/`

# Ignore VS Code settings

`.vscode/`

# Ignore build folder

`build/`

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### **Step 3: Save the file.**

Now Git will ignore these files whenever you run:

`git status`

They will not appear as untracked files.

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## How to Implement .gitignore

### Example Project

MyProject/

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├─ src/

├─ node\_modules/

├─ .env

├─ app.log

├─ README.md

└─ .gitignore

### .gitignore file

node\_modules/

.env

\*.log

Now:

- node\_modules/ → Ignored
  - .env → Ignored
  - app.log → Ignored
  - README.md → Tracked by Git
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## Important Note

If a file is **already being tracked by Git**, adding it to `.gitignore` **will not stop Git from tracking it**.

You first need to remove it from Git's tracking:

```
git rm --cached filename
```

Example:

```
git rm --cached .env
```

Then commit the changes:

```
git commit -m "Stop tracking .env"
```

Now `.env` will be ignored because it is listed in `.gitignore`.